

LED AND LUMINAIRE TESTING CAN'T BE TAKEN LIGHTLY

Standards and tests are essential to evaluate the quality and longevity of LEDs and luminaires. So test and measurement solutions have to keep pace with the evolving LED lighting standards



Photometry Testing Lab (Courtesy: Goldwyn)

■ SHWETA DHADIWAL BAID

Though the market for light-emitting diodes (LEDs) is poised to grow and replace compact fluorescent lamps (CFLs) and high-intensity discharge (HID) lamps, LED lighting has its own share of challenges. At recently held *Executive Congress on LED Lighting* in Delhi, Saurabh Kumar, secretary, Bureau of Energy Efficiency (BEE), said that the lack of technical specifications for LEDs is the biggest constraining factor in India.

There are certain international bodies working on standards for LED luminaires. "Commission Internationale de L'Eclairage (CIE), Illuminating Engineering Society (IES) and American

National Standards Institute (ANSI) standards are universally accepted to test and characterise optical and electrical parameters of LEDs," shares Atulya Sahay, director-Photonics Marketing Company. "For consumers, Energy Star is voluntary energy-efficiency labeling that is awarded to a product that meets strict energy guidelines set by US Department of Energy (DOE) and US Environmental Protection Agency (EPA)."

"Government's concrete policies to advocate green products will encourage manufacturers to produce and consumers to purchase eco-friendly products," adds Carol Chiu, marketing manager, GlacialTech.

Standards are continuously evolving

to improve the quality and longevity of the products. "Test and measurement solutions are required to function according to the new standards. The most recent standards IES LM79 and IES LM80 were not in existence six years ago, so the older testing solutions cannot be used to perform the new tests. T&M solutions are evolving to catch up with the evolving standards," explains Vikrant Mahajan, applications engineer, Labsphere.

There is a huge variation from one manufacturer to the other in terms of specifications of a product. Mahajan shares, "T&M providers work with the manufacturers to find what they want to test and build. Normally, 70 per cent of the requirements can be satisfied by commercial off-the-shelf equipment, but the rest 30 per cent is permutation and combination through customised solutions."

The three main tests performed on LEDs and luminaires are electrical, thermal and optical tests. All these tests are done to get the specifications of the luminaire comprising the LED chip, LED driver and packaging. Nimish Buddhadev, business head-India, Tayo Industrial Group, says, "The biggest challenge is to test its life, consistency and then the driver quality."

ETO characterisation

Electrical, thermal and optical (ETO) characteristics of LEDs help product designers to estimate the performance of their LED-based modules and products. The tests are performed on LED chips, LED drivers, the package and